

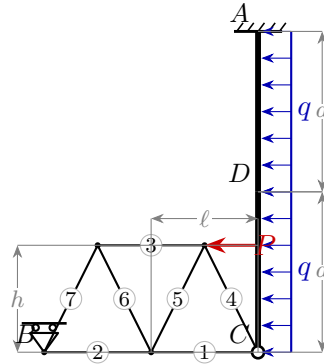
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Exam **Group A**

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Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

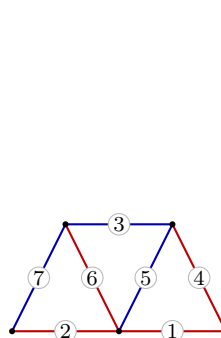
A : $R_x = 50$, $R_y = -10$, $M = 210$ B : $R_x = 0$, $R_y = 10$

C : $H_x = -20$, $H_y = 10$

3. Truss member forces

Member	Force S [kN]	Type
1	15	tension
2	5	tension
3	-10	compression
4	11.18	tension
5	-11.18	compression
6	11.18	tension
7	-11.18	compression

red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams